

Proofs by Case Analysis in Isabelle

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

1. Context & Intuition

Revision of Previous Topics:

- Isabelle interface, terms, and theorem statements.
- Forward reasoning (FW), Backward reasoning (BW).
- Proofs by substitution (**subst**).
- Mathematical induction for natural numbers (**nat**).

Case Analysis Intuition:

- When proving a property for all numbers/lists, different cases may require distinct reasoning paths.

- **Example (Monus / Subtraction on nat):**

$$3 - 1 = 2, \quad 3 - 2 = 1, \quad 3 - 3 = 0, \quad 3 - 4 = 0$$

In Isabelle, $x - y = 0$ whenever $x \leq y$. Thus, proving statements with subtraction often requires a case split on $x \geq y$ vs. $x < y$.

2. Worked Example: Proving $x - y \leq x$

Theorem: $x - y \leq x$ (Induction on x)

Base Case ($x = 0$):

- **goal** :: $0 - y \leq 0$
- **Proof steps:**
 1. $0 - y = 0$ (by *subst lem1*: $0 - a = 0$)
 2. $0 \leq 0$ (by *lem2*: $a \leq a$)
- finally $0 - y \leq 0$ ✓

3. Step Case ($x \rightarrow \text{Suc } x$) with Case Analysis

Induction Hypothesis (I.H.): $x - y \leq x$

Goal: $(\text{Suc } x) - y \leq \text{Suc } x$

Proof by Case Analysis on $x \geq y$:

Case 1 ($x \geq y$):

1. $(\text{Suc } x) - y = \text{Suc } (x - y)$ (by *lem1* [Case 1 *assump*])
2. $(\text{Suc } x) - y \leq \text{Suc } (x - y)$ (from 1, *lem3*)
3. $\dots \leq \text{Suc } x$ (using *lem4* [OF I.H.]) ✓

Case 2 ($x \not\geq y$, i.e. $x < y$):

1. $x < y$ (from *lem5* [OF Case 2 *assump*])
2. $\text{Suc } x \leq y$ (from 1)
3. $(\text{Suc } x) - y = 0$ (from *lem2* [OF 2])
4. $0 \leq \text{Suc } x$ (from *lem6*)
5. $(\text{Suc } x) - y \leq \text{Suc } x$ (using 4, *subst* 3) ✓

Auxiliary Lemmas Used in Proof

- **lem1:** $a \geq b \implies (\text{Suc } a) - b = \text{Suc } (a - b)$
- **lem2:** $a \leq b \implies a - b = 0$
- **lem3:** $a \leq a$
- **lem4:** $a \leq b \implies \text{Suc } a \leq \text{Suc } b$
- **lem5:** $a \not\geq b \implies a < b$
- **lem6:** $0 \leq a$